Day 7

① Bubble Sort

[5, 1, 4, 2, 8]

5 > 1 swap [1 5 4 2 8]
5 > 4 swap [1 4 5 2 8]
5 > 2 swap [1 4 2 5 8]
5 < 8 no swap

4 > 2 → swap [1 2 4 5 8]

next pass no swap, stop

final [1, 2, 4, 5, 8]

② Selection Sort

[64 25 12 22 11]

smallest = 11

[11 25 12 22 64]

Page No.
Date

smallest = 12

[11 12 25 22 64]

smallest = 22

[11 12 22 25 64]

final = [11, 12, 22, 25, 64] Time $O(n^2)$

③ Insertion Sort

[12 11 13 5 6]

take one element (key) & insert into its correct position

key 11	11	12	13	5	6
key 13	12	12	13	5	6
key 5	5	11	12	13	6
key 6	5	6	11	12	13

④ Merge Sort

[38 27 43 3]

[38 27] [43 3]

again [38][27] [43][3]

merge [38] + [27] → [27, 38]
[43] + [3] → [3, 43]

[27 38] [3 43]

[3 27 38 43] O(nlogn)

⑤ Quick sort

[10 80 30 90 40 50 70]

pivot = 70

elements < 70 goes left and elements > 70 goes to right

10 30 40 50 70 80 90

Average O(nlogn)
worst $O(n^2)$

⑥ Heap sort

Array [4 10 3 5 1]

Date

MaxHeap [10 5 3 4 1]

10 moved to end

5 4 3 1 10

again move largest

4 1 3 5 10

repeat this process

final: [1 3 4 5 10] O(nlogn)